① Calcula la potencia que soporta la resistencia y si se quemará o no

$R = 2.2K\Omega \quad 1/4\,W$

$I = \dfrac{15V}{2.2K} = 6.81mA \qquad P = 6.81mA(15)$

$P = 0.1\,W \qquad 0.1\,W < 1/4\,W$

$\therefore$ No se quemará la resistencia

$R = 22\Omega \quad 1/4\,W$

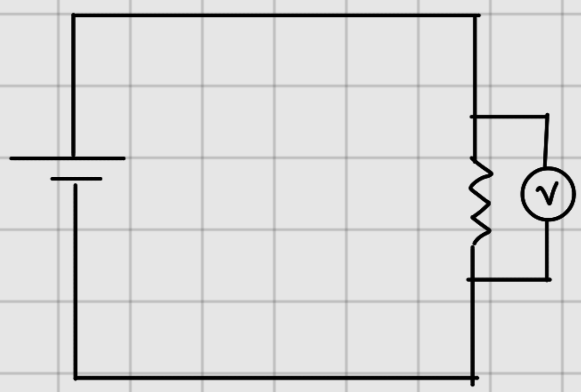
$I = \dfrac{15V}{22\Omega} = 0.68A \qquad P = 0.68A(15)$

$P = 10.22\,W \qquad 10.22\,W > 1/4\,W$

$\therefore$ Si se quemará la resistencia

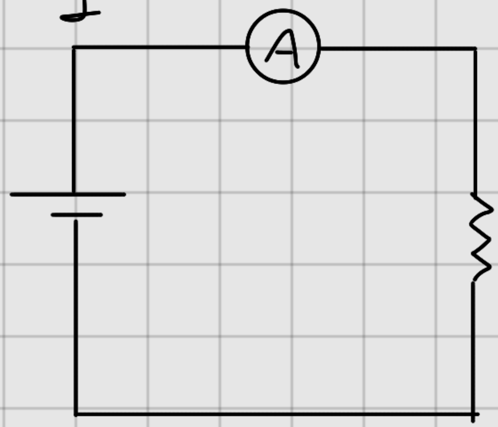
② Como se mide voltaje y corriente

V



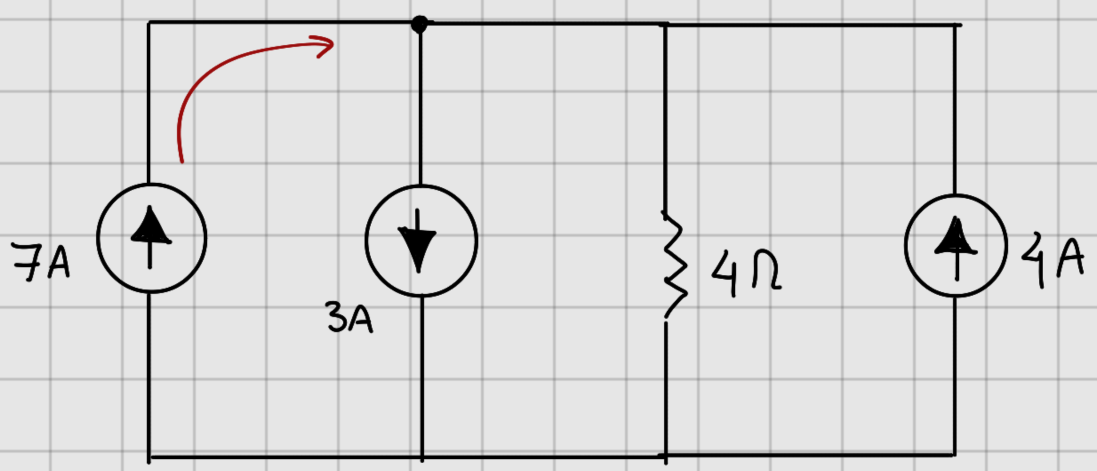
$R = \infty\ \Omega$

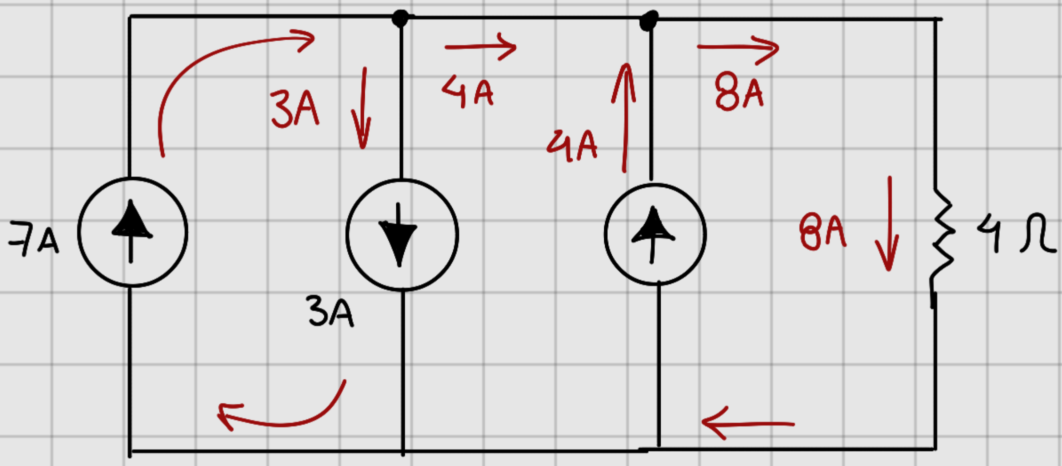
I



$R = 0\ \Omega$

③ Resuelve el circuito por leyes de corriente de Kirchoff. Encuentra la potencia en la R.



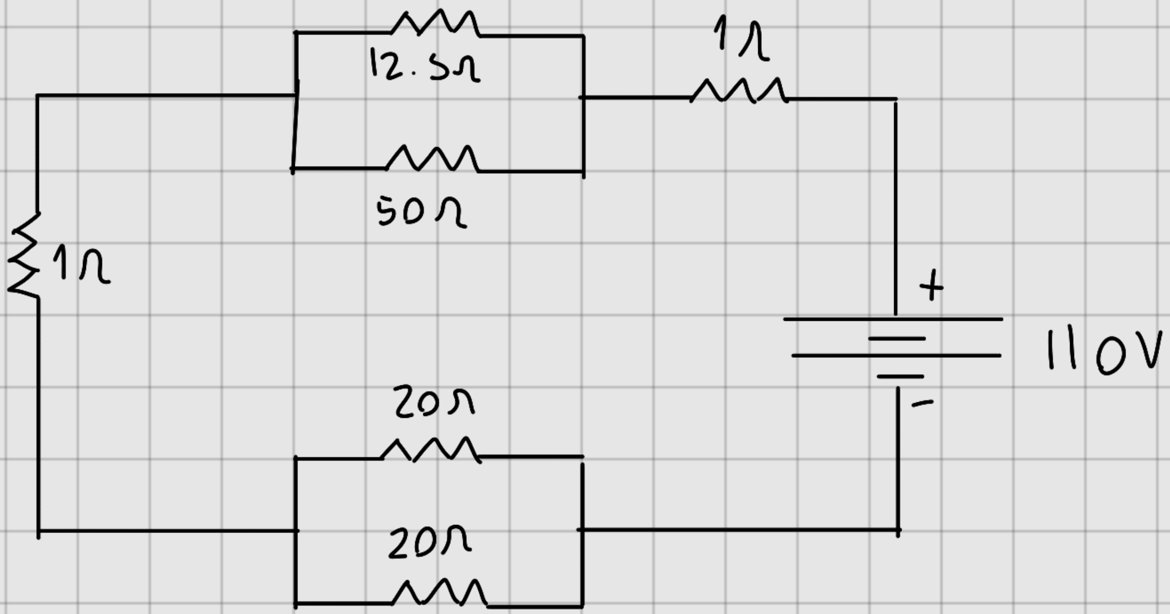


$$V = IR = 8A(4\Omega) = 32\,V$$

$$P = IV = 32V(8A) = 256\,W$$

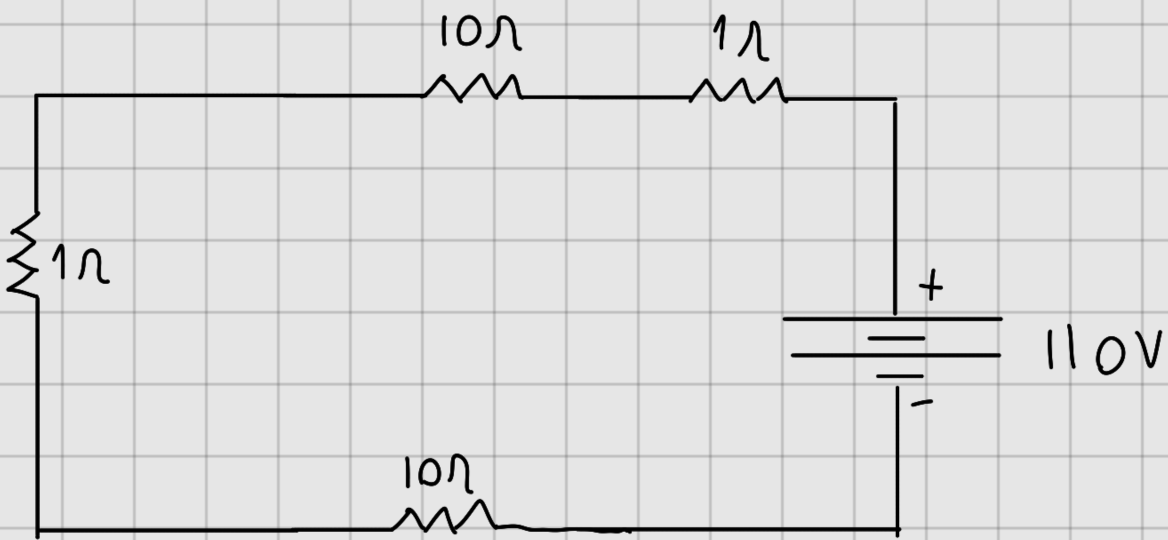
$$\underline{P = 256\,W}$$

④ Resuelve y encuentra la potencia total del circuito



$$R_1 = \frac{1}{\frac{1}{12.5\Omega} + \frac{1}{50\Omega}} = \frac{1}{0.08 + 0.02} = \frac{1}{0.1} = 10\Omega$$

$$R_2 = 10\Omega$$



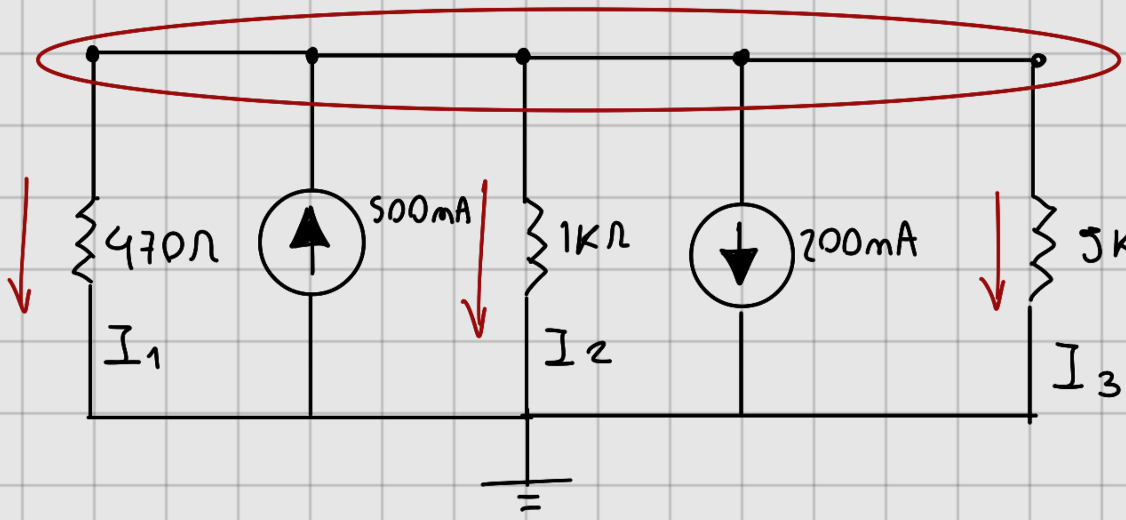
$$R = 1\Omega + 10\Omega + 10\Omega + 1\Omega = 22\Omega$$

$$I = \frac{V}{R_T} = \frac{110V}{22\Omega} = 5A$$

$$P = IV = 5A \times 110V = 550W$$

$$\underline{\underline{P = 550\ W}}$$

⑤ LCK. Halla I_1, I_2 y I_3.



$$-I_{R_1} + I_1 - I_2 - I_{R_2} - I_{R_3} = 0$$

$$V\left(\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_1}\right) = (I_1 - I_2)$$

$$V = \frac{(I_1 - I_2)}{\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_1}} = \frac{300\,mA}{0.033\,S} = 90V$$

$$I_1 = \frac{90}{470\Omega} = 0.1914A$$

$$I_2 = \frac{90}{1000\Omega} = 0.09A$$

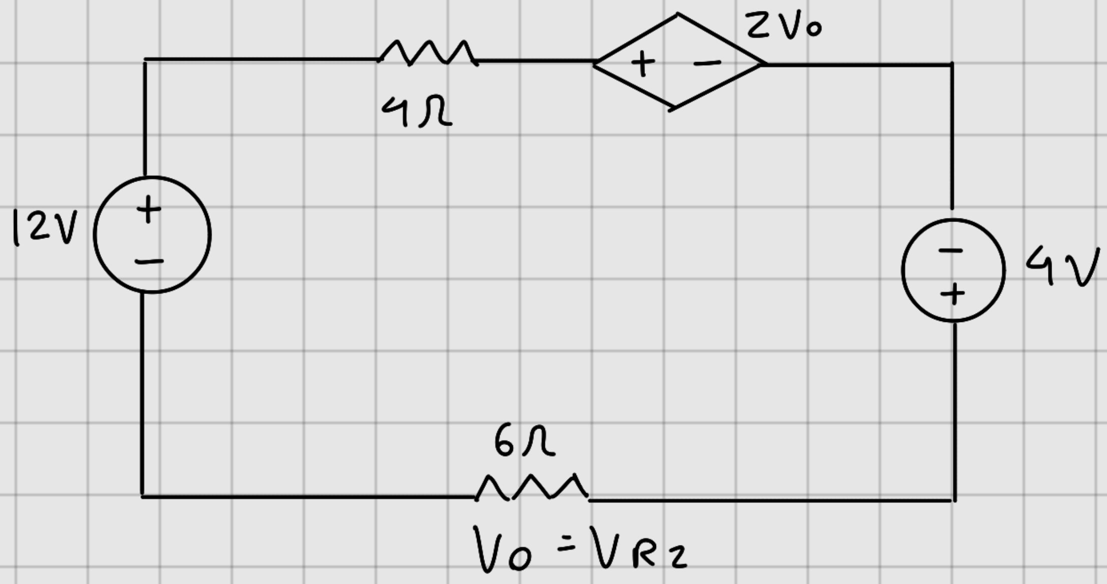
$$I_3 = \frac{90}{5000\Omega} = 0.018A$$

$I_1 = 0.1914A$

$I_2 = 0.090A$

$I_3 = 0.018A$

⑥ LVK. Halle v, V_o y I_T



$$V_1 - V_{R1} - 2V_o + V_2 - V_{R2} = 0$$

$$V_1 - IR_1 - 2IR_2 + V_2 - I \cdot R_2 = 0$$

$$V_1 + V_2 - I\left(R_1 + 2R_2 + R_3\right) = 0$$

$$\rightarrow I(R_1 + 3R_2) = (V_1 + V_2)$$

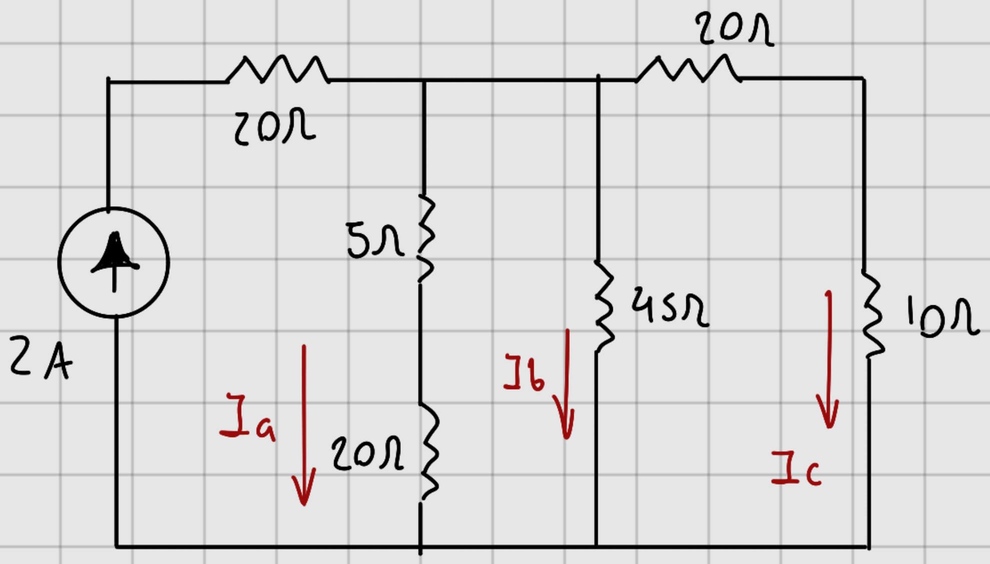
$$I = \frac{V_1 + V_2}{R_1 + 3R_2} = \frac{16V}{22\Omega} = 0.7272A$$

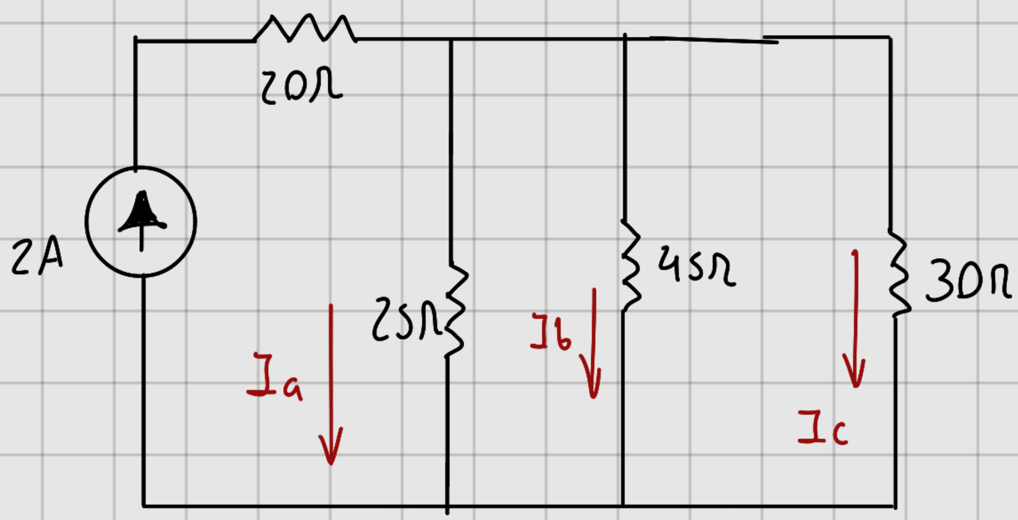
$$I \cdot R = V = (0.72727A)(6\Omega) = 4.36V$$

$$v = 2V_o = 2(4.36V) = 8.72V$$

$$I = 0.7272A,\ v = 8.72V,\ V_o = 4.36V$$

⑦ Resuelve por Divisor de corriente y encuentre I_a, I_b, I_c y la P en $R = 10\,\Omega$





$I_T = 2A$

$I_x = \dfrac{R_T\, I_T}{R_x} =$

$$R_{eq} = \frac{1}{\frac{1}{25} + \frac{1}{45} + \frac{1}{30}} = 10.46\,\Omega$$

$$I_a = \frac{10.46\cancel{\Omega}(2A)}{25\cancel{\Omega}} = 0.8372\,A$$

$$I_b = \frac{10.46\cancel{\Omega}(2A)}{45\cancel{\Omega}} = 0.465\,A$$

$$I_c = \frac{10.46\cancel{\Omega}(2A)}{30\cancel{\Omega}} = 0.6978\,A$$

$$P_{R10\Omega} = I_c(10\Omega) \times I_c$$

$$= 6.978 \times 0.6978 = 4.86\,W$$

$I_a = 0.8372\,A$

$I_b = 0.465\,A$

$I_c = 0.6978\,A$

$P_{R10\Omega} = 4.86\,W$